

Homework 7

40 points

1. Given, $f(x) = (x^2 + 2x + 3)^5$. Find the first and second order derivative of the given function. **3 + 3 = 6 points**

2. Given $y = \sin (x^2 + 3x)$. Find $\frac{dy}{dx}$ **4 points**

3. Find all values of x for which the tangent to $f(x) = x^4 - 4x^3 + 3$ will be horizontal. **5 points**

4. Find f_x and f_y for the following: **5*2 = 10 points**

- $f(x, y) = x^2y + 3xy^2$
- $f(x, y) = \frac{xy}{x^2+y^2}$ (for $(x,y) \neq (0,0)$)

5. Compute all second order partial derivatives of $f(x, y) = x^3y^2 - 5x^2y$ **4*2**
= 8 points

6. Given $f(x, y, z) = x^2 + y^2 + z^2$

- Find the gradient vector at point $(1, 2, 2)$ **3 points**
- Find the directional derivative at that point in the direction of the vector $(1, 1, 1)$ **2 points**
- Find the maximum rate of change. **2 points**